Valentin Michels

Bonn, Germany | vmichel1@uni-bonn.de | | | | |

Applied Geophysicist with 6+ years of experience in signal processing, inversion theory, and Python-based data analysis. Expert in designing automated workflows to transform terabyte-scale geoscientific datasets into actionable, data-driven decisions.

Skills

Software	Python (Scipy, Scikit-learn, PyTorch, Polars, Pandas), PostgreSQL, Git, Linux, Bash, R, HPC
Geophysics	Forward Mod. & Inv. (PyGIMLi, SimPEG, CRTomo), Methods (IP, ERT, Gravity, Magnetics, Seismics)
Data Science	Signal Processing (EM Correction, Wavelet & Timeseries Analysis), Automation (SQLAlchemy, Cron)
Geospatial & Viz	QGIS, Geopandas, Rasterio, Grid creation (Gmsh, ParaView), Visualization (Plotly Dash, PyVista)
Soft Skills	Project Management, Mentoring & Supervision, German (Native), English (Fluent), French

Research & Professional Experience

Research Scientist, *University of Bonn (Excellence Cluster Phenorob)* **Mar. 2020 – present**

- Developed automated IoT data streams for remote spectral electrical impedance tomography (sEIT) monitoring systems, reducing data latency from weeks to near real-time and enabling continuous in-situ geophysical imaging of crop roots.
- Implemented correction procedures to mitigate high-frequency EM effects in sEIT surveys, enabling scaling the method from lab to field scale for precise quantification of root traits.
- Eliminated manual processing bottlenecks and significantly increased data throughput by engineering automated inversion & processing pipelines for terabyte-scale complex resistivity data.
- Engineered interactive dashboards for multi-modal data fusion, accelerating the joint interpretation of geophysical models and environmental time-series data.
- Mentored student thesis projects and led academic training, providing scientific and technical guidance on state-of-the-art geophysical theory and application for reproducible research.

Visiting Researcher, *Lawrence Berkeley National Laboratory* Oct. 2022 – Feb. 2023

- Secured competitive funding for an international collaboration with Yuxin Wu's Lab at LBNL to quantify structural root traits in high-throughput sEIT experiments.
- Enhanced resolution and data quality of geoelectrical and induced polarization surveys through algorithmically optimized measurement schemes.

Early Research & Industry Experience May 2016 – Apr. 2019

- **M.Sc Intern:** Integrated numerical simulations of fluid flow with experimental electrical and self-potential data to characterize fluid flow and transport processes in porous media.
- **B.Sc Intern:** Quantified saturation-dependent soil properties using a custom-engineered spectral induced polarization measurement cell to deliver high-precision laboratory datasets.
- **Intern** *(KÜHN Geoconsulting)*: Executed fieldwork including soil/rock/water sampling, soil compaction monitoring, and drill core preparation for geotechnical site assessments.

Education

PhD in Geophysics (summa cum laude), *University of Bonn* Mar. 2020 – Oct. 2026

MSc (Grade 1.3) & BSc (Grade 1.8) in Geophysics/Geosciences, *University of Bonn* Oct. 2012 – Apr. 2019

Awards & Community Engagement

Grants: DAAD Mobility Stipend (2025); Santander International Research Grant (2022)

Leadership: Organized Early-Career Researcher Assembly at the German IP Workshop (2021)

Volunteering: Environmental conservation with Conservation Volunteers Australia (2012)